

NLP

Week#02

NLP

- NLP = building **computer programs** to analyze, understand and generate **human language - either spoken or written** (informal)
- Natural language processing (NLP) is a subfield of computer science and [artificial intelligence \(AI\)](#) that uses [machine learning](#) to enable computers to understand and communicate with human language.
- NLP enables computers and digital devices to recognize, understand and generate text and speech by combining computational linguistics, the rule-based modeling of human language together with statistical modeling, machine learning and [deep learning](#).

NLP Applications

Core

- Text Classification
- Information Extraction
- Conversational Agent
- Information Retrieval
- Question Answering Systems

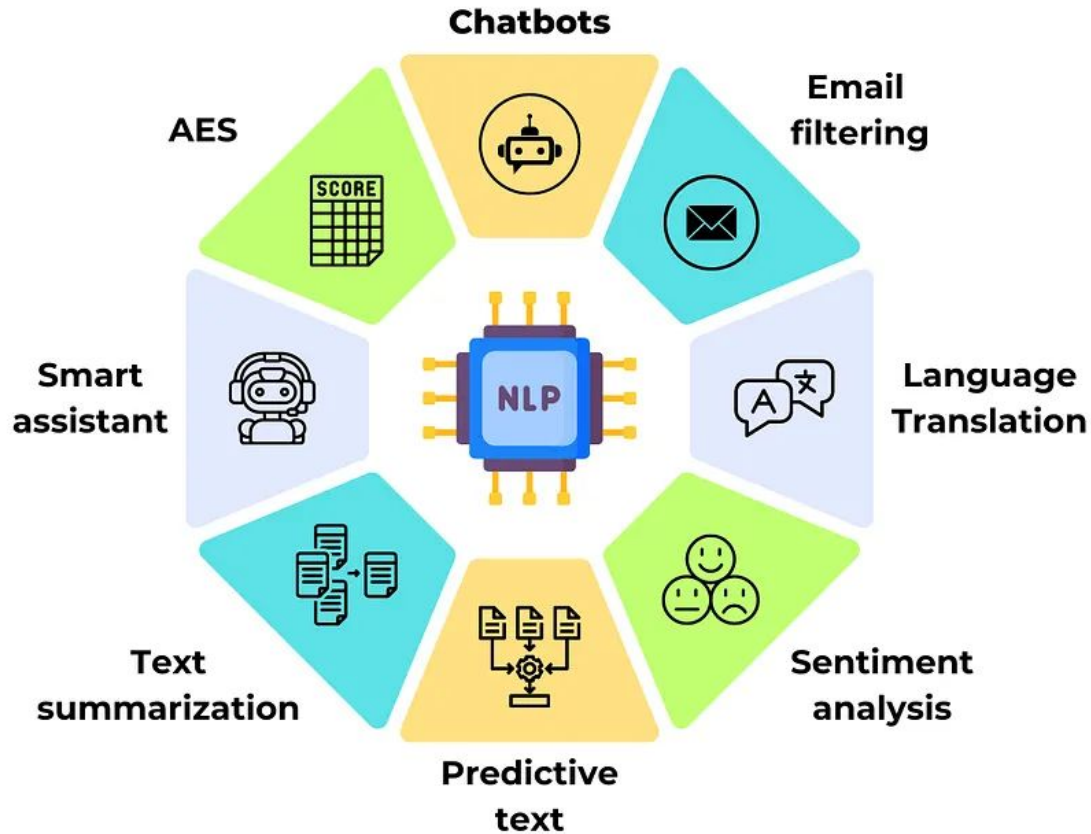
General App

- Spam Classification
- Calendar Event Extraction
- Personal Assistants
- Search Engines
- Jeopardy!

Industry Specific

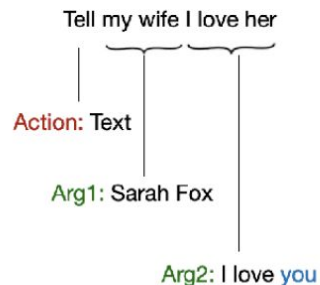
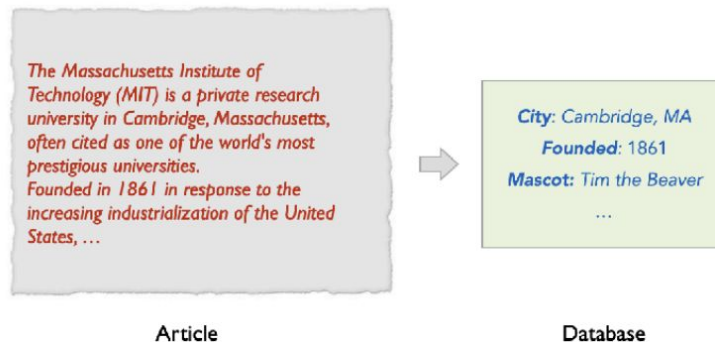
- Social Media Analysis
- Retail Catalog Extraction
- Health Records Analysis
- Financial Analysis
- Legal Entity Extraction

Applications of Natural Language Processing



NLP subfields

- Machine translation
- Question answering
- Information extraction
- Text summarization
- Dialogue
- Semantic parsing
- Syntactic parsing
- Language grounding
-

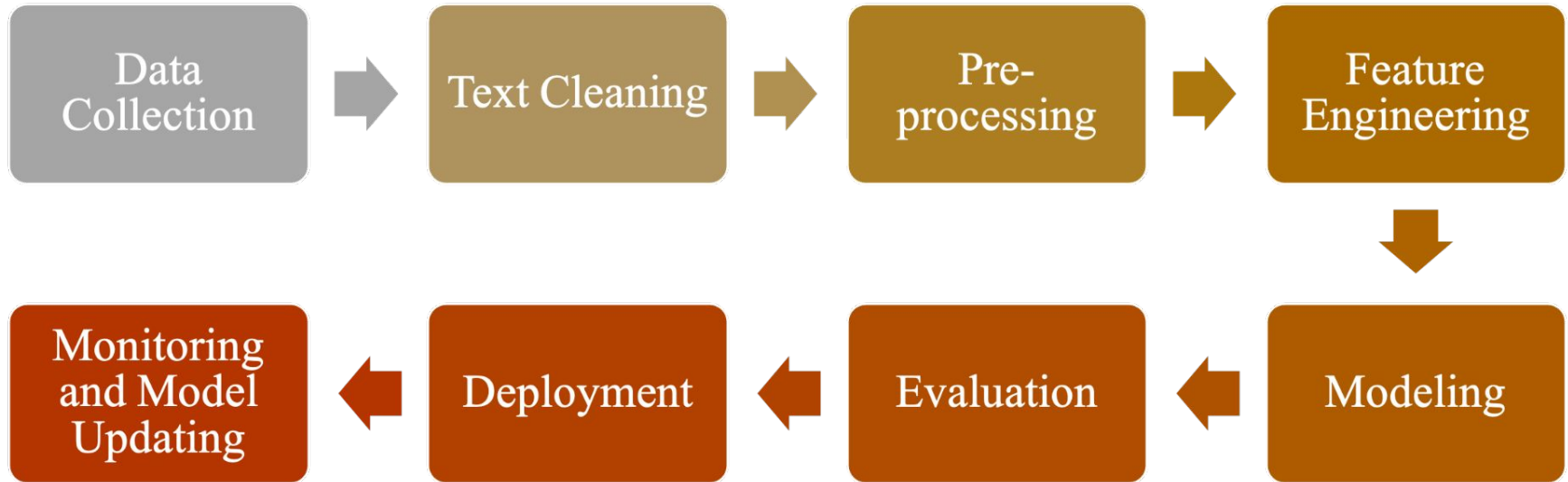


Approaches

1. Heuristic Approach
2. Machine Learning Approach
3. Deep Learning Approach
4. Cloud API

Challenges in NLP

NLP Pipeline / Workflow



Data Collection

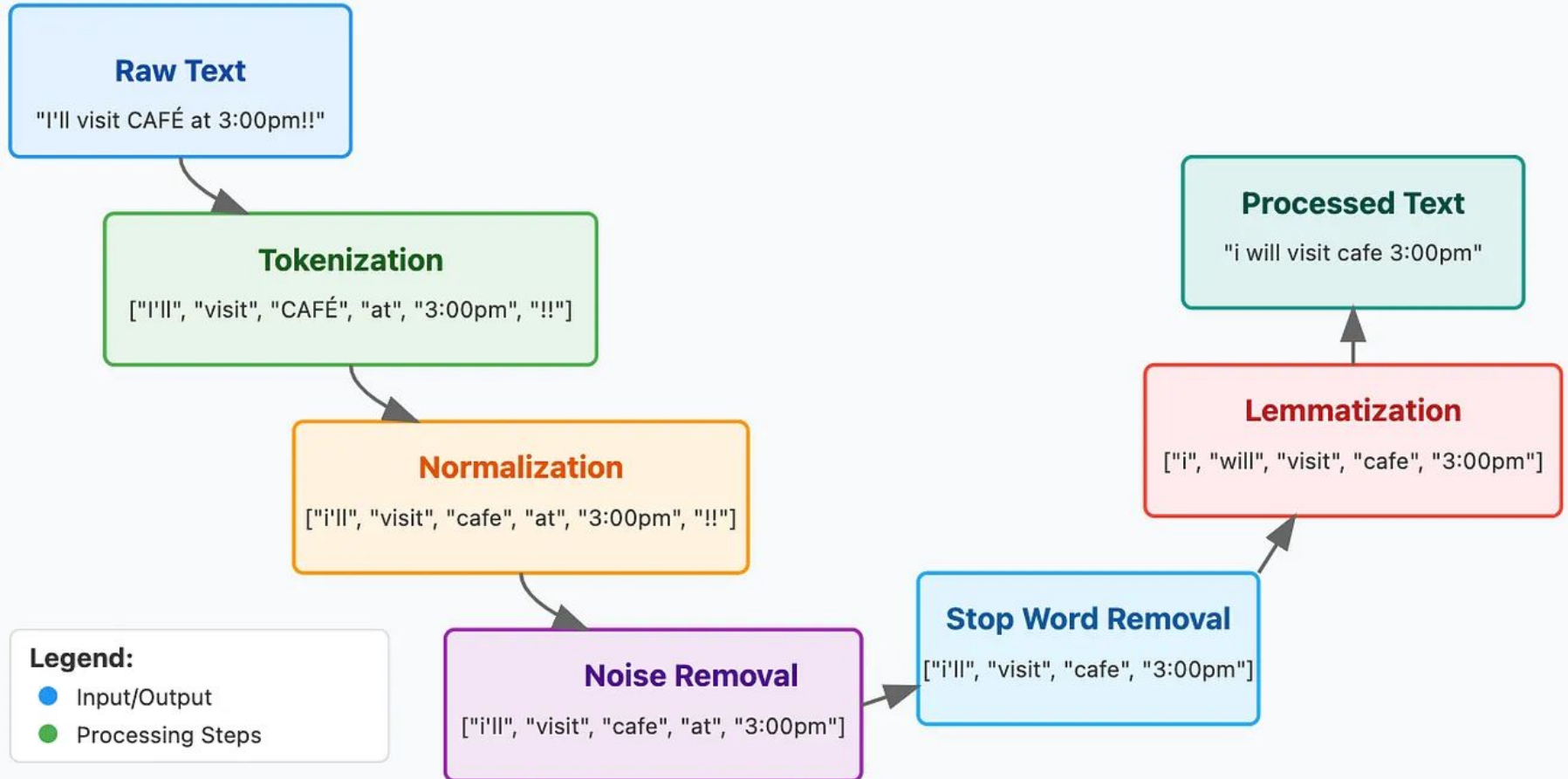
There are three possible situations happen.

1. Data Available Already (Ideal Setting)
 - a. Complete datasets with rich labels & annotations
2. Less-than-Ideal Scenarios
 - a. Limited labeled data
 - b. Labels from regex or heuristics
 - c. Public datasets (Google Dataset Search, Kaggle)
 - d. Scraped data
 - e. Product intervention → curated/biased data
3. Data Augmentation (NLP)
 - Expands training data by creating diverse, syntactically similar samples
 - Strategies:
 - Synonym replacement
 - Related word replacement (association metrics)
 - Back translation
 - Entity replacement
 - Adding noise (spelling errors, random word insertions)

Text Processing in NLP

Text Normalization	Text Tokenization	Text Enrichment
• HTML tags • Stemming • Lemmatization • Contractions • Accented Characters • Special Characters • Redundant Whitespaces • Stopwords	• words • sentences	• POS Tagging • Chunking • Parsing

Text Preprocessing Pipeline



Text Processing in NLP

Important Reminders for Preprocessing

- Not all steps are necessary
- These steps are NOT sequential
- These steps are task-dependent and language-dependent
- Goals
 - Text Normalization
 - Text Tokenization
 - Text Enrichment/Annotation

Text Normalization

The objective of text normalization is to clean up the text by removing unnecessary and irrelevant components.

Feature Engineering

Feature Engineering means converting text data to numerical data.

Why do we need to convert it to numerical values ?

There are multiple techniques to convert text to numerical vectors.

1. One Hot Encoder
2. Bag Of Word(BOW)
3. n-grams
4. Tf-Idf
5. Word2vec

train_X

'This is good',

'This is bad'

'This is awesome'

Fit

CountVectorizer

word_index

{'this':0,
'is':1,
'good':2,
'bad':3,
'awesome':4}

Features

This	is	good	bad	awesome
1	1	1	0	0
1	1	0	1	0
1	1	0	0	1

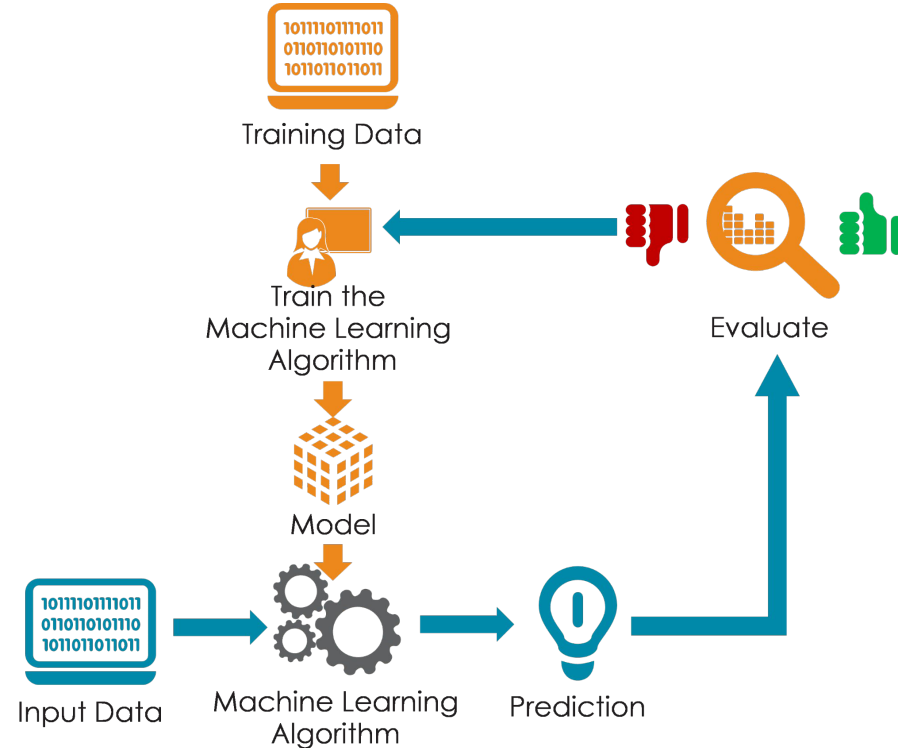
Modelling/Model Building

Approaches to building model –

1. Heuristic Approach
2. Machine Learning Approach
3. Deep Learning Approach
4. Cloud API

Which approach to use ?

1. Amount of data
2. Nature of the problem.



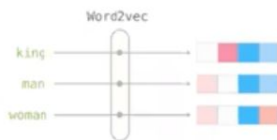
Model Evaluation

In this evaluation, we use multiple metrics to check our model such as Accuracy, Recall, Confusion Metrics, Perplexity, etc.

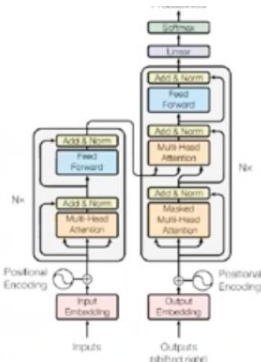
Modern NLP

A brief history

1990s-2010's
[The NLP Winter]



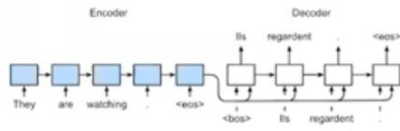
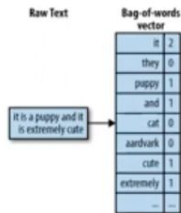
2014
[Recurrence & Machine Translation]



2018
[BERT – One Model to Rule Them All]



2013
[First Forays into Deep Learning]



2017
[“Attention” is All We Need]



2023
[The “Zero-Shot” World?]

The field of AI is experiencing unprecedented progress that will disrupt every industry; this is driven by 3 main factors.



Advancements in **machine learning** and open source **toolsets**

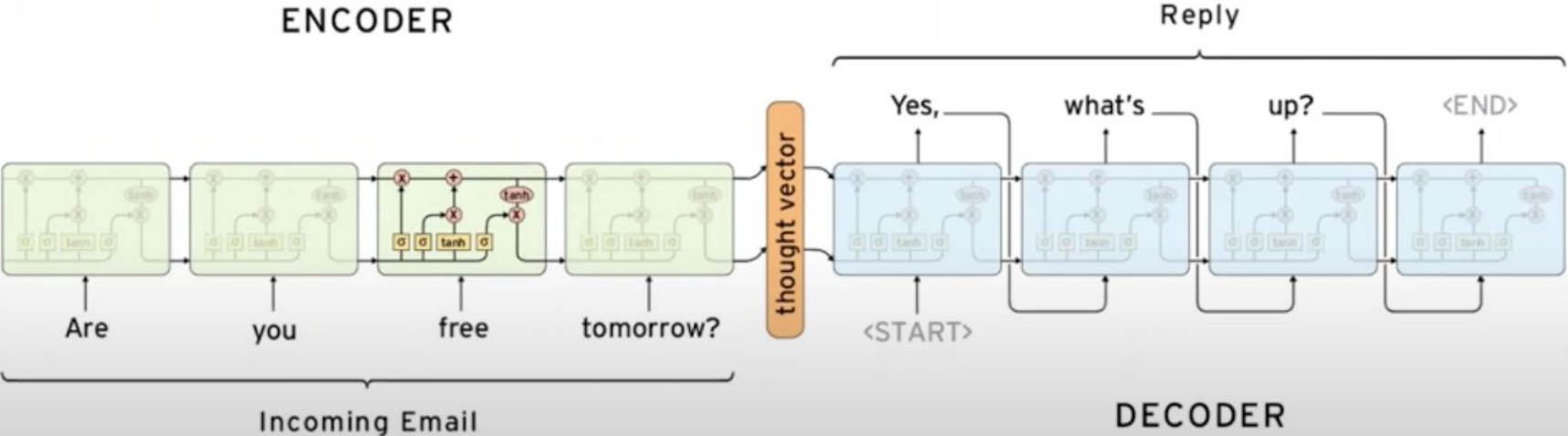


Enormous **growth of data**

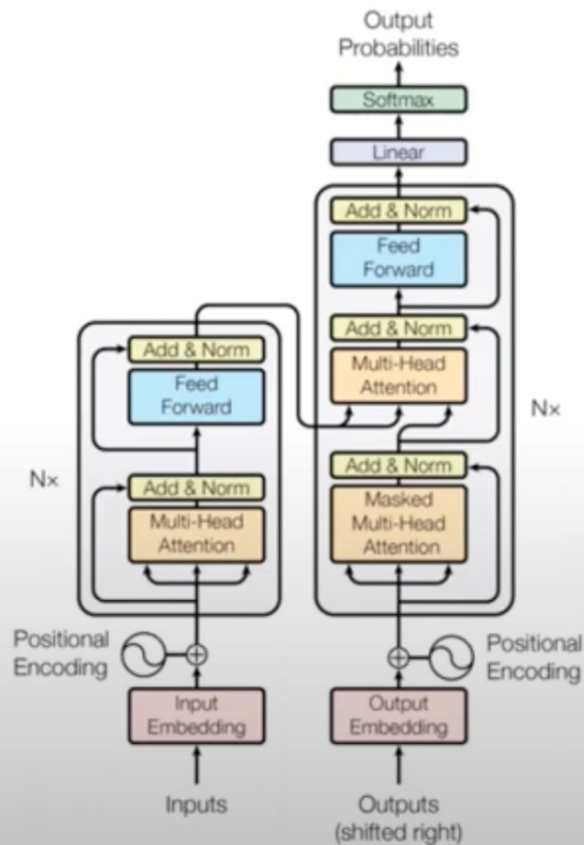
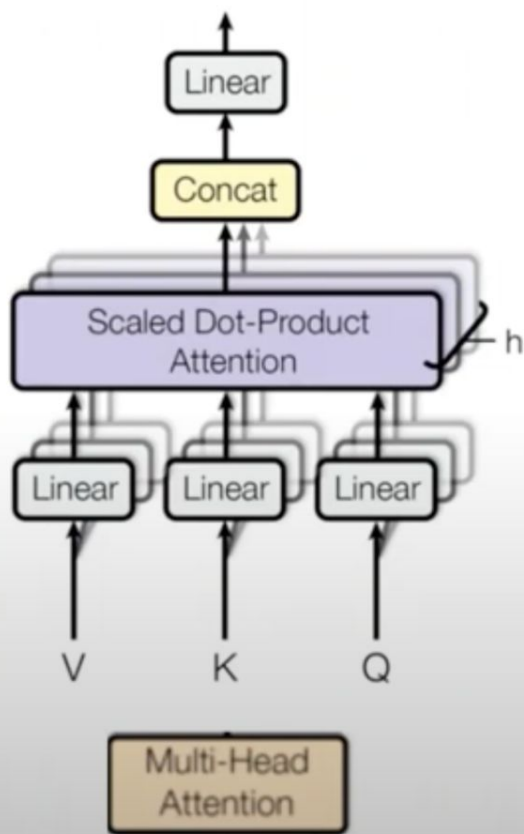


Advancement in hardware and optimization, **making better predictions cheaper**

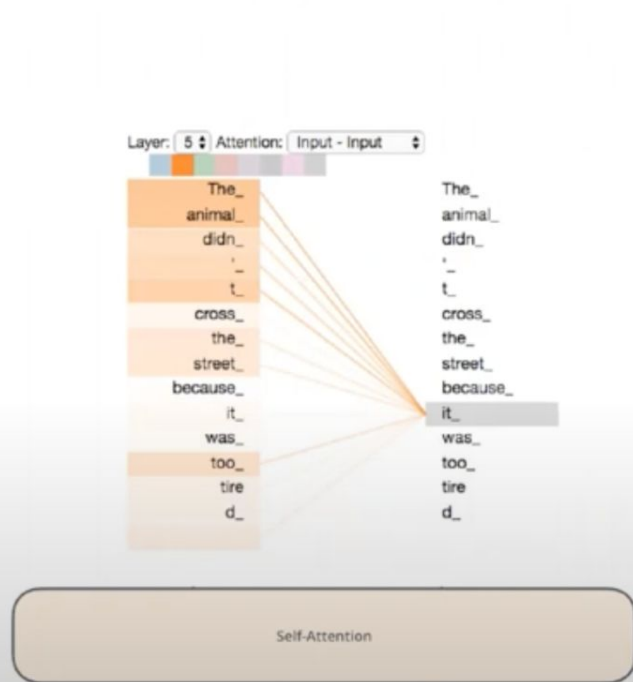
The evolution of architectures – from LSTM's to Transformers



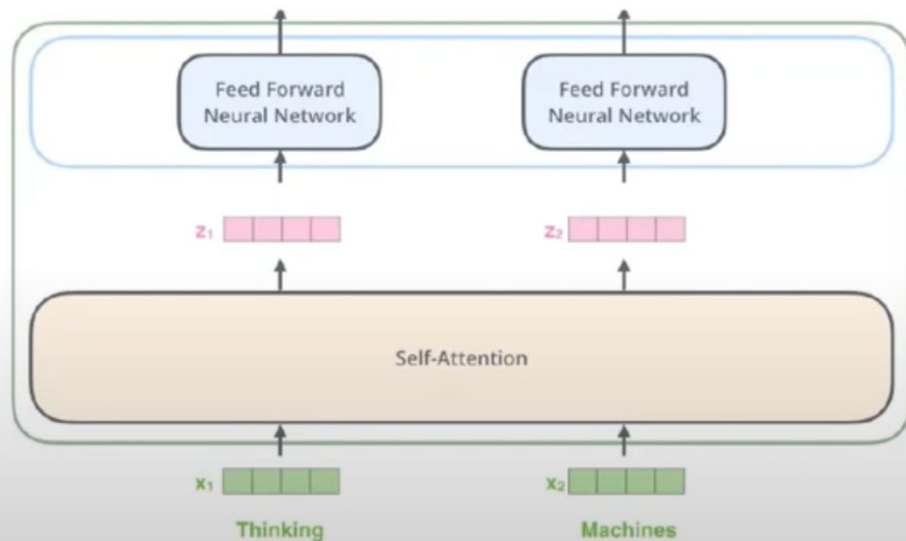
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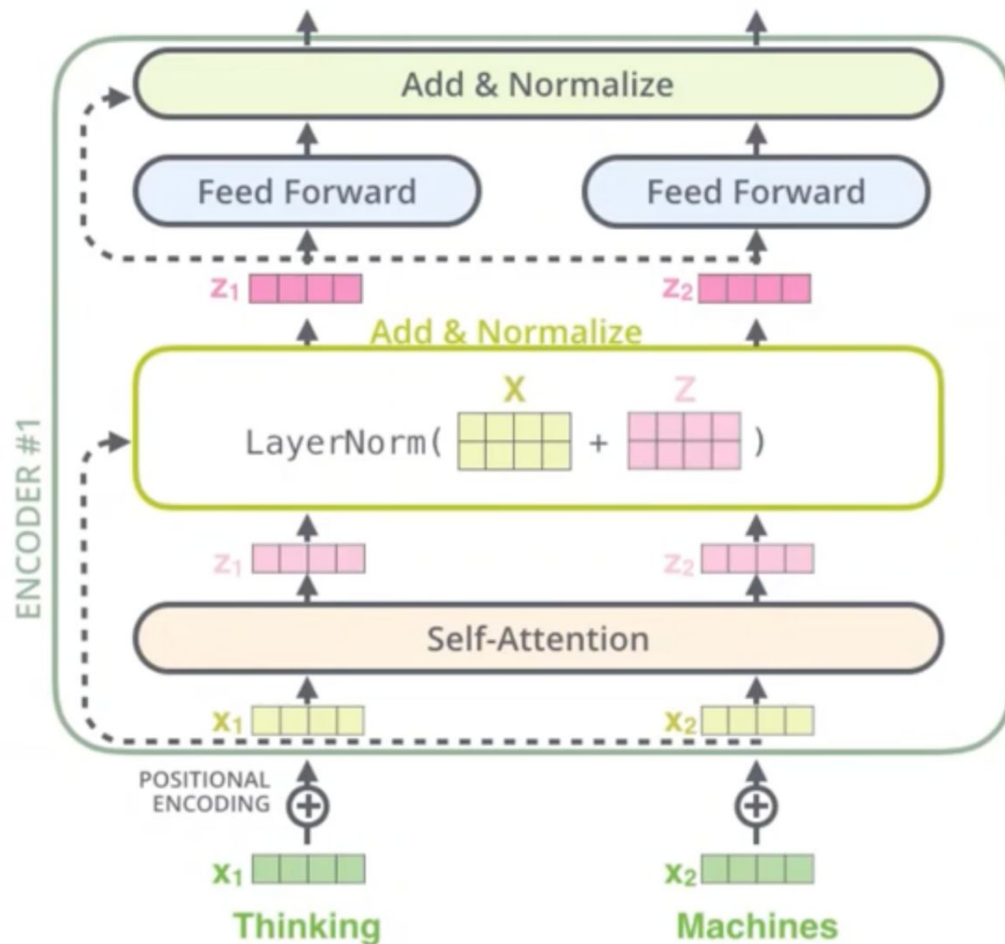
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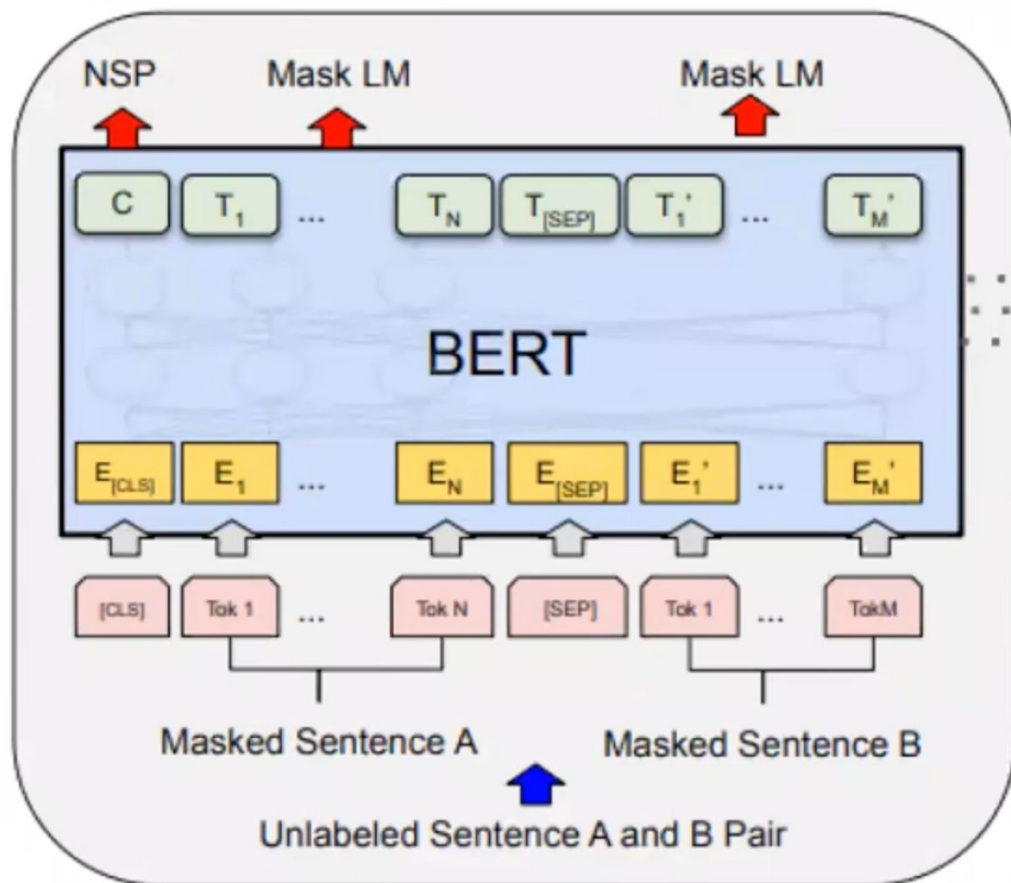
ENCODER #1



The evolution of architectures – from LSTM's to Transformers



Datasets for Creating Language Models



Reading Task

<https://www.analyticsvidhya.com/blog/2022/06/an-end-to-end-guide-on-nlp-pipeline/>

Read this article and implement it

Read this paper :

“BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding“

<https://arxiv.org/abs/1810.04805>